Class Notes on os modules

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◇ 1. What is the OS Module?

The **os module** in Python provides a way to **interact with the operating system.**

👉 It allows your Python program to:

- Work with files and directories
- Get system information
- Execute system commands
- Handle paths (cross-platform)

◇ 2. Why Use the OS Module?

Without os, Python is limited to basic operations. With it, you can:

✓ Create/Delete folders
✓ Navigate directories
✓ Access environment variables
✓ Run shell commands
✓ Build platform-independent programs

◇ 3. Importing the Module

```
import os
```

◇ 4. Working with Directories

📌 Get Current Directory

```
import os
print(os.getcwd())
```

👉 Output: Current working directory

📌 Change Directory

```
os.chdir("D:/Python")
```

📌 **List Files/Folders**

```
print(os.listdir())
```

👉 Lists all files in current directory

📌 **Create a Directory**

```
os.mkdir("NewFolder")
```

📌 **Create Multiple Directories**

```
os.makedirs("A/B/C")
```

📌 **Remove Directory**

```
os.rmdir("NewFolder")
```

◇ 5. Working with Files

📌 **Rename File**

```
os.rename("old.txt", "new.txt")
```

📌 **Delete File**

```
os.remove("file.txt")
```

◇ 6. Path Handling (Very Important)

Use os.path for safe path operations.

📌 **Join Paths**

```
path = os.path.join("folder", "file.txt")
print(path)
```

📌 **Check File Exists**

```
print(os.path.exists("file.txt"))
```

📌 **Check File or Directory**

```python
print(os.path.isfile("file.txt"))
print(os.path.isdir("folder"))
```

📌 **Get File Size**

```python
print(os.path.getsize("file.txt"))
```

◇ 7. Environment Variables

📌 **Get Environment Variable**

```python
print(os.environ.get("PATH"))
```

📌 **Set Environment Variable**

```python
os.environ["MY_VAR"] = "Hello"
```

◇ 8. Running System Commands

📌 **Execute Command**

```python
os.system("dir")    # Windows
os.system("ls")     # Linux/Mac
```

◇ 9. Process Management

📌 **Get Process ID**

```python
print(os.getpid())
```

📌 **Get User Info**

```python
print(os.getlogin())
```

◇ 10. Practical Example (Mini Program)

🔸 **Create Folder + File + Check Info**

```python
import os
folder = "TestFolder"
# Create folder
if not os.path.exists(folder):
    os.mkdir(folder)
```

```python
# Create file
file_path = os.path.join(folder, "sample.txt")
with open(file_path, "w") as f:
    f.write("Hello OS Module")
# Display info
print("File exists:",
os.path.exists(file_path))
print("File size:", os.path.getsize(file_path))
```

◇ 11. Important Functions Summary

Function	Purpose
getcwd()	Current directory
chdir()	Change directory
listdir()	List files
mkdir()	Create folder
remove()	Delete file
rename()	Rename file
path.exists()	Check existence

◇ 12. Advantages of OS Module

✓ Platform-independent code

✓ Powerful file handling

✓ System-level access

✓ Useful for automation scripts